Fiscal Year:	FY 2017	**Task Last Updated:**	FY 08/01/2016
PI Name:	Plawsky, Joel Sc.D.		
Project Title:	Constrained Vapor Bubble Flight Experiment II – Binary Mixtures		
Division Name:	Physical Sciences		
Program/Discipline:	FLUID PHYSICS		
Program/Discipline--Element/Subdiscipline:	FLUID PHYSICS--Fluid physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	plawsky@rpi.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	518-276-6049
Organization Name:	Rensselaer Polytechnic Institute		
PI Address 1:	Department of Chemical and Biological Engineering		
PI Address 2:	110 Eighth Street, RI-122		
PI Web Page:	http://www.rpi.edu/		
City:	Troy	**State:**	NY
Zip Code:	12180-3522	**Congressional District:**	20
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	NOT AVAILABLE
Start Date:	10/01/2013	**End Date:**	09/30/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	2	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA GRC
Contact Monitor:	Chao, David	**Contact Phone:**	216-433-8320
Contact Email:	david.f.chao@nasa.gov		
Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Wayner, Peter (Rensselaer Polytechnic Institute)		
Grant/Contract No.:	NNX13AQ78G		
Performance Goal No.:			
Performance Goal Text:			

The thermophysical principles underlying change-of-phase heat transfer systems controlled by interfacial phenomena under microgravity conditions are not well understood. As a result, related passive engineering systems like heat pipes have not been optimized and it is unclear whether recent advances in the design of these systems based on ground-based research using fluid mixtures can be translated to microgravity systems. Herein, we propose basic experimental and theoretical studies of the nonisothermal constrained vapor bubble (CVB-2) using a mixture of pentane and isohexane under microgravity conditions to help remedy this undesirable situation. This proposed study is multi-faceted: 1) it is a basic scientific study in interfacial phenomena, microgravity fluid physics, and thermodynamics; 2) it is a basic study in thermal and mass transport; and 3) it is a study of a passive heat exchanger. Although the basic scientific Facets (1) and (2) are emphasized, the research naturally leads to the development of a passive heat exchanger optimized for microgravity conditions.
The proposed optical study of a vapor bubble constrained in a transparent glass cell under nonisothermal conditions will

Task Description:	increase the basic understanding of transport systems controlled by interfacial phenomena. The pressure gradient field will be obtained optically whereas the temperature field will be obtained using thermal sensors. Due to the sensitivity of systems of the proposed relatively large size to gravity, to small temperature and pressure gradients, and to changes in fluid composition, these transport systems need to be studied under microgravity conditions. An "axisymmetric" system with a small Bond number is required. Direct extensions to the microgravity environment first investigated in CVB-1 are proposed. The augmented Young-Laplace equation and the Kelvin-Clapeyron model will be used to help understand and quantify how the system operates. The immediate macroscopic objectives of CVB-2 are to determine the stability, the fluid flow characteristics, the average heat transfer coefficient in the evaporator and condenser sections, and the heat conductance of the CVB as a function of the heat input using the new fluid mixture. The immediate microscopic objective is to determine the detailed characteristics of the transport processes in the curved liquid film. The local conditions under which cavitation and instability occur with the formation of a dry region will be determined as a function of heat flux, film thickness, and stress. High fluxes and relatively large systems with small capillary pressures will be emphasized. The data using the pentane/isohexane mixture will be compared with the pure pentane data obtained in CVB-1 and ground-based data on identical fluid mixtures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This work addresses fundamental questions regarding phase-change heat and mass transfer operations that are becoming increasingly important in a wide variety of fields ranging from energy conversion, to microelectronics cooling, micro-electromechanical systems (MEMS) devices, self-assembly, and space flight. Of particular concern to NASA are microchannel and heat pipe systems used for thermal management and perhaps separation processes. Such systems are attractive for long-term space flights because they generally require no moving parts and are potentially maintenance free. The ability to control transport processes in an extended meniscus will be of great utility both on Earth and in space, whether we are looking for enhanced heat transfer, coating, or self-assembly applications. Some of our preliminary ground-based experiments concerning the microscopic evaporating meniscus, performed as part of the existing CVB-1 flight experiment program, have indicated that using a fluid mixture, with the minority component being the less volatile, higher surface tension liquid, offers the opportunity to control fluid flow and heat transfer in an evaporating meniscus and thus enhance the performance of heat pipes. The surface concentration gradient will also enhance axial fluid flow in the corners. The extension of the CVB-1 experiment to mixtures allows us to isolate the Marangoni effects from the effects of gravity and begin to understand how to exploit Marangoni flows for improving the performance of heat pipes in space and on Earth.
Task Progress:	During the course of this year we have been vigorously working on analyzing the data from both CVB2 and CVB1 flight experiments. We have focused most recently on 50x data to see what we could recover from those sets of experiments and to see what differences we could see at that magnification level between CVB-1 and 2. We have been writing papers, and presenting the information at as many conferences as we could reasonably attend. The highlights are listed below. 1. We spent a considerable amount of time regressing and analyzing the data from the high heat input experiments run aboard the International Space Station (ISS) in late 2010 using the 30 and 40 mm module CVB1 experiments on the ISS and developing enhanced thermal models to determine the internal heat transfer rates within the cuvette. This has resulted in a publication in the International Journal of Multiphase Flow. Much more analysis work remains but we have discovered the limits of the Marangoni phenomenon and what may be driving it. 2. We have analyzed the low resolution (surveillance and 10x) images associated with CVB-2 and have developed mathematical models tying the images and the temperature profile to the fluid composition and the performance of the evaporator section of the heat pipe. This has resulted in a publication in the International Journal of Heat and Mass Transfer. 3. We have analyzed the effect of cooler temperature on the evaporator performance of CVB-2. Even though the cooler is located over 30 mm away from the heater, altering the cooler temperature had a great effect on the heat pipe's performance and the composition near the heater end. This work resulted in a publication in Experimental Thermal and Fluid Science. 4. We have recently uncovered similar data from CVB-1. This is currently being analyzed but shows that the cooler temperature even for the pure fluid, exerts a tremendous influence on the processes occurring at the heater end. We are still analyzing the data and will be submitting a paper on this topic in 2017. 5. The high heat input data on CVB-1 has led us to discover two new phenomena. The first is wholesale condensation at the heater end of the device which we believe serves as the source for fluid accumulation at that end. We believe it is due to disjoining pressure effects and if so, would validate one of the earliest, scientific justifications for running CVB-1. The high heat data has also led to a self-organizing, reverse jet that forms on the flat surfaces of the CVB (CVB-1) near the heater end. This also drives fluid from regions where evaporation is small (near the heater) to where it is large (near the central drop). A combination of condensation, evaporation, and Marangoni flow, we believe, leads to the behavior. These two topics are the subject of at least one paper that will be submitted by September 2016. 6. Our study of 50x images from CVB-2 and how the meniscus behaves in that device is the subject of a paper to be submitted by September 2016. 7. We submitted a proposal to NASA to use the Physical Sciences Informatics database to continue analyzing and modeling results from CVB1. The PI on that proposal was Vladimir Ajaev of Southern Methodist University. 8. We submitted a proposal to the National Science Foundation/Center for the Advancement of Science in Space (NSF/CASIS) fluid dynamics call. The goal was to expand on CVB-2 and fly a nearly 50:50 mixture of pentane and isohexane on the ISS. In this region, odd surface tension effects were observed in ground-based experiments, and very turbulent heat transfer performance was also observed. The project has in-kind support from Intel and also Jay Mann and Bill Meyer.

	9. We presented results at 3 meetings/seminars with two more schedule for 2016. 10. Akshay Kundan defended his thesis in the Spring of 2016 and will graduating in the summer.
Bibliography Type:	Description: (Last Updated: 10/14/2025)
Articles in Peer-reviewed Journals	NguyenTTT, Kundan A, Wayner PC Jr, Plawsky JL, Chao DF, Sicker RJ. "The effect of an ideal fluid mixture on the evaporator performance of a heat pipe in microgravity." International Journal of Heat and Mass Transfer. 2016 Apr;95:765-72. http://dx.doi.org/10.1016/j.ijheatmasstransfer.2015.12.032 , Apr-2016
Articles in Peer-reviewed Journals	Kundan A, Nguyen TTT, Plawsky JL, Wayner PC Jr, Chao DF, Sicker RJ. "Arresting the phenomenon of heater flooding in a wickless heat pipe in microgravity." International Journal of Multiphase Flow. 2016 Jun;82:65-73. http://dx.doi.org/10.1016/j.ijmultiphaseflow.2016.02.001 , Jun-2016
Articles in Peer-reviewed Journals	Nguyen TTT, Kundan A, Wayner PC Jr, Plawsky JL, Chao DF, Sicker RJ. "Effects of cooling temperature on heat pipe evaporator performance using an ideal fluid mixture in microgravity." Experimental Thermal and Fluid Science. 2016 Jul;75:108-17. http://dx.doi.org/10.1016/j.expthermflusci.2016.01.016 , Jul-2016
Dissertations and Theses	Kundan A. "The Constrained Vapor Bubble Heat Pipe Experiment: Understanding the Role of Interfacial Flow." Dissertation, Rensselaer Polytechnic Institute, August, 2016. , Aug-2016